MODULE 3 — FEATURE EXTRACTION

1. Introduction to Feature Extraction

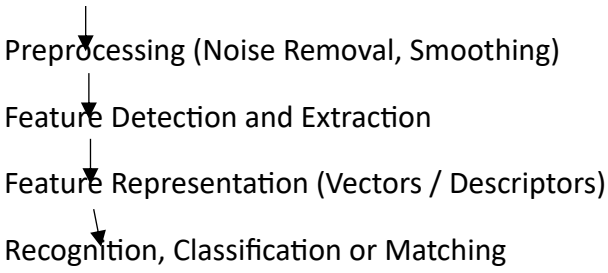
Definition

Feature extraction is the process of identifying and representing important characteristics of an image, such as edges, corners, lines, textures, shapes, and local patterns. These features help a computer vision system recognize objects, classify images, and understand the scene.

Instead of processing every pixel directly, feature extraction converts an image into a set of meaningful measurements.

Examples: edges of buildings, corners of a chessboard, texture of fabric, and keypoints on a human face.

Block diagram



Working

1. Acquire an image using a camera or scanner.
2. Preprocess the image to reduce noise and improve quality.
3. Detect important image structures.
4. Calculate numerical features or descriptors.
5. Use the features for classification, matching, tracking, or segmentation.

Advantages

- Reduces the amount of information required for recognition.
- Helps identify objects under varying conditions.
- Improves matching and classification.
- Supports automated visual inspection.

Disadvantages

- Noise and illumination changes can affect features.

- Some features may be lost during preprocessing.
- Different objects may produce similar features.
- Complex scenes require more computation.

Applications

Face recognition, medical image analysis, autonomous driving, object tracking, fingerprint matching, and industrial quality control.

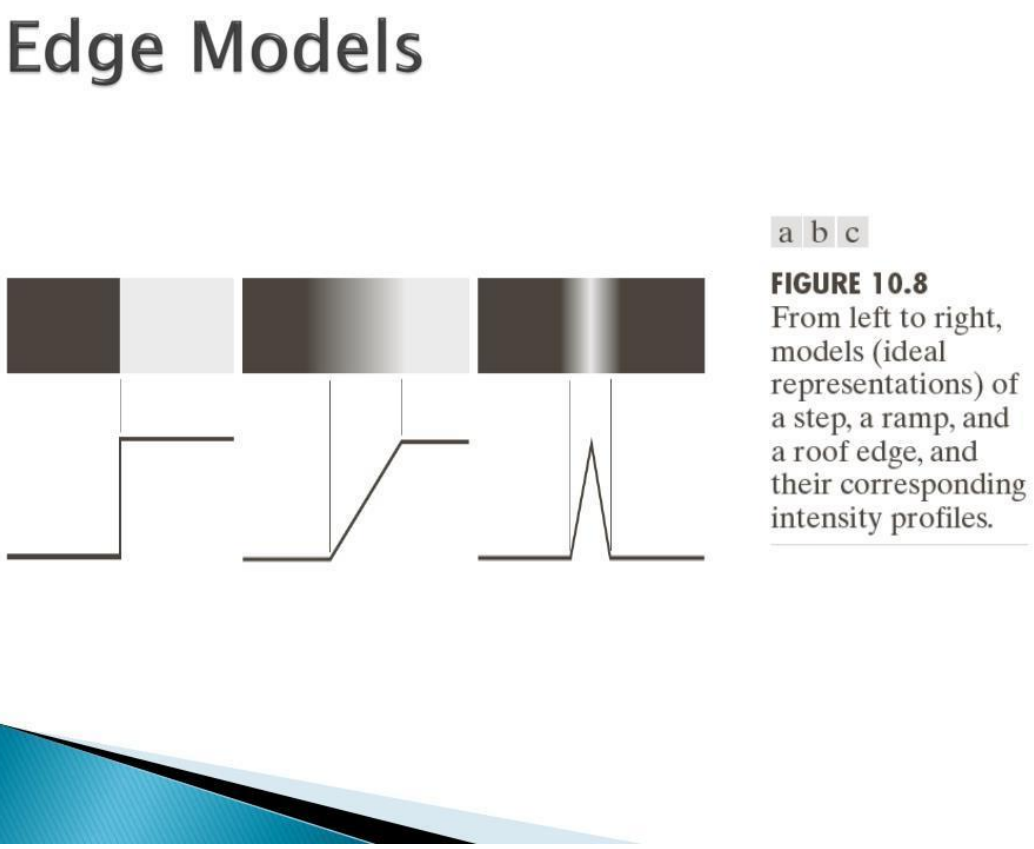
2. Edge Detection

Definition

An edge is a location in an image where the intensity or colour changes significantly. Edge detection identifies these locations to reveal object boundaries and structural details.

Edges commonly occur at boundaries between an object and its background, between two different objects, or where surface orientation or illumination changes.

Types of edges



1. Step edge

Intensity changes suddenly from one level to another. It commonly represents a clear object boundary.

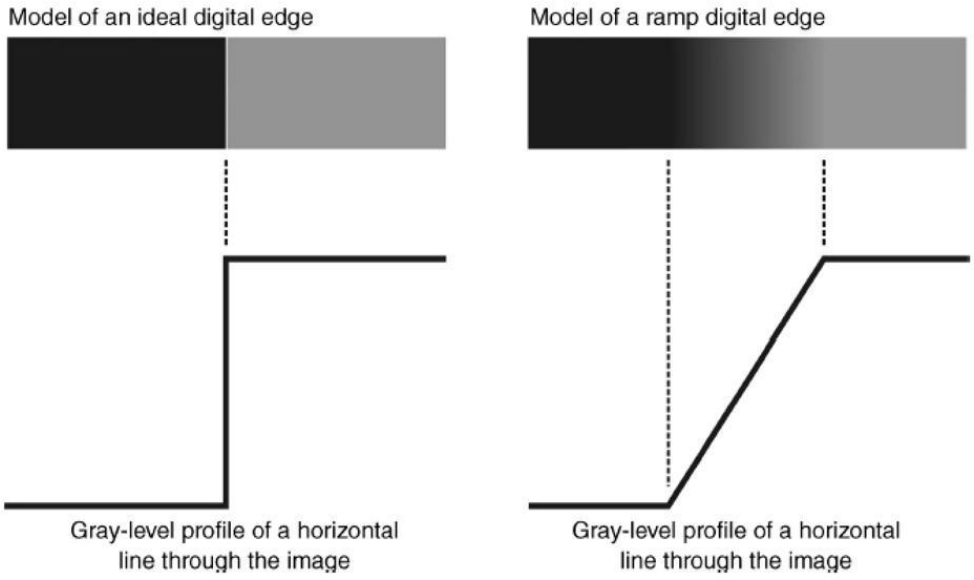
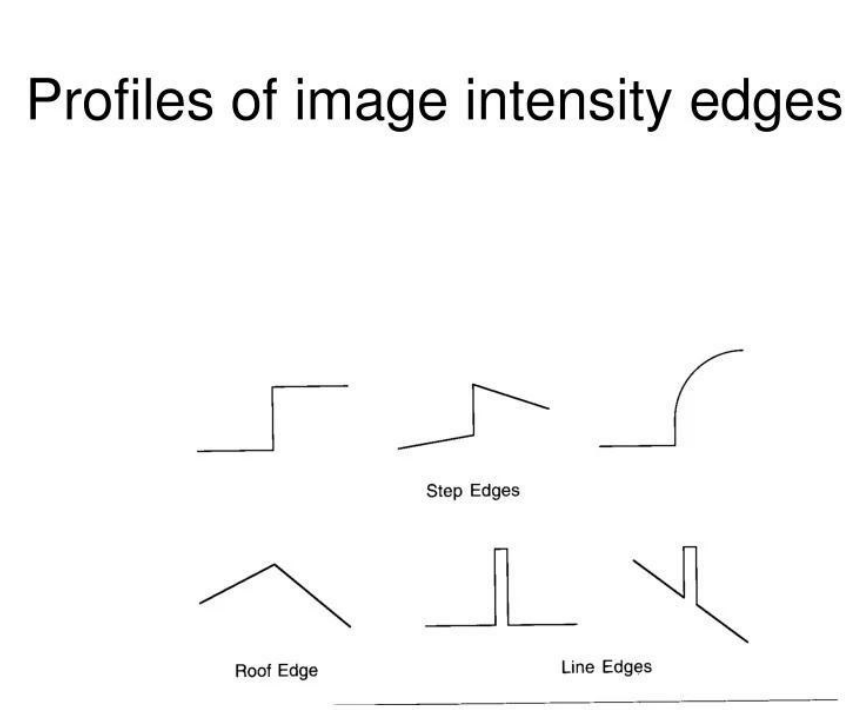


FIGURE 14.1 Ideal and ramp edges: (a) ideal edge on a digital image and corresponding profile along a horizontal line; (b) ramp edge and corresponding profile.

[Ch. 14, Practical Image and Video Processing Using MATLAB by Oge Marques]

2. Ramp edge

Intensity changes gradually over a region, often because of blur or gradual illumination changes.



3. Roof edge

Intensity rises and then falls, producing a narrow ridge-like feature. It may represent thin lines or ridges.

Mathematical principle

An edge can be detected using the image intensity gradient:

$$ \nabla I=\begin{bmatrix} \frac{\partial I}{\partial x}\\[3pt] \frac{\partial I}{\partial y} \end{bmatrix} $$

Gradient magnitude:

$$ G=\sqrt{G_x^2+G_y^2} $$

Gradient direction:

$$ \theta=\tan^{-1}\left(\frac{G_y}{G_x}\right) $$

A large gradient magnitude indicates a possible edge.

General working

1. Convert the image to grayscale.
2. Reduce noise using a smoothing filter.

3. Calculate intensity changes.
4. Apply a threshold or edge-selection rule.
5. Produce a binary or edge-strength image.

Advantages

- Identifies object boundaries.
- Reduces an image to its important structural features.
- Helps object detection and segmentation.

Disadvantages

- Noise can produce false edges.
- Weak edges may be missed.
- Edge detection alone does not identify the object.

Applications

Road-lane detection, medical boundaries, industrial inspection, shape analysis, and image segmentation.

3. Canny Edge Detection

Definition

Canny edge detection is a multi-stage edge detection technique developed by John Canny. It aims to produce edges that are well localized, have low error rates, and contain minimal multiple responses to a single edge.

Block diagram

Input Image

↓

Gaussian Smoothing

↓

Gradient Calculation

↓

Non-Maximum Suppression

↓

Double Thresholding

↓

Edge Tracking by Hysteresis

↓

Final Edge Map

Working steps

Step 1: Gaussian smoothing

Removes noise before calculating gradients.

$$ G(x,y)=\frac{1}{2\pi\sigma^2}e^{-\frac{x^2+y^2}{2\sigma^2}} $$

The image is convolved with the Gaussian kernel.

Step 2: Gradient calculation

Sobel operators estimate horizontal and vertical intensity changes.

$$ M=\sqrt{G_x^2+G_y^2} $$

Step 3: Non-maximum suppression

Keeps only local maxima along the gradient direction, making edges thin.

Step 4: Double thresholding

Two thresholds are used:

- Strong edge: $M\geq T_H$
- Weak edge: $T_L\leq M<T_H$
- Non-edge: $M<T_L$

Here, T_H is the high threshold and T_L is the low threshold.

Step 5: Edge tracking by hysteresis

Strong edges are retained. Weak edges are retained when connected to strong edges; isolated weak responses are removed.

Algorithm

1. Read the input image.
2. Convert it to grayscale.
3. Apply Gaussian smoothing.
4. Compute gradient magnitude and direction.
5. Apply non-maximum suppression.
6. Apply high and low thresholds.
7. Track weak edges connected to strong edges.
8. Output the final edge map.

Advantages

- Good edge localization.

- Low false-edge response compared with many simple methods.
- Produces thin edges.
- Handles noise better because of smoothing and hysteresis.

Disadvantages

- Requires several processing stages.
- Performance depends on Gaussian scale and thresholds.
- Computationally more expensive than simple gradient operators.
- Fine details may disappear with excessive smoothing.

Applications

Road and lane detection, object boundary extraction, medical imaging, industrial inspection, and shape recognition.

4. LOG Edge Detection — Laplacian of Gaussian

Definition

The Laplacian of Gaussian (LoG) is an edge detection method that first smooths an image using a Gaussian filter and then applies the Laplacian operator. Edges are identified using zero crossings of the resulting image.

It is useful when noise must be reduced before detecting rapid intensity changes.

Principle

The Gaussian filter is:

$$ G(x,y)=\frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} $$

The Laplacian operator is:

$$ \nabla^2 I= \frac{\partial^2 I}{\partial x^2}+ \frac{\partial^2 I}{\partial y^2} $$

Therefore,

$$ LoG=\nabla^2(G*I) $$

where I is the image, G is the Gaussian kernel, and $*$ represents convolution.

Block diagram

Input Image

↓

Gaussian Filter — Noise Reduction

↓

Laplacian — Second Derivative

↓

Zero-Crossing Detection

Edge Map

Working

1. Read and convert the image to grayscale.
2. Apply Gaussian smoothing to suppress noise.
3. Apply the Laplacian second-derivative operator.
4. Detect positions where the response changes sign.
5. Reject insignificant changes using a threshold, if required.
6. Mark the remaining zero crossings as edges.

Algorithm

Input: Image I

1. Convert I to grayscale.

2. Smooth I using Gaussian filter G.

3. Compute L = Laplacian(G * I).

4. Examine neighbouring pixels in L.

5. Detect sign changes across neighbours.

6. Retain significant zero crossings.

Output: Edge map

Advantages

- Detects edges in multiple orientations.
- Gaussian smoothing reduces some noise effects.
- Zero crossings can reveal fine image structures.
- A single operator can respond to edges without explicitly choosing a gradient direction.

Disadvantages

- Sensitive to noise if smoothing is insufficient.
- Zero crossings may produce false or double edges.
- Edge localization can be less direct than with gradient methods.

- Results depend strongly on Gaussian scale.

Applications

Medical image analysis, surface inspection, boundary detection, texture analysis, and scientific image processing.

5. DOG Edge Detection — Difference of Gaussians

Definition

Difference of Gaussians (DoG) is an image-processing technique that subtracts two Gaussian-blurred versions of the same image, using different standard deviations. It highlights structures at a particular spatial scale and is commonly used for feature detection.

Unlike LoG, DoG approximates the Laplacian of Gaussian using the difference between two Gaussian filters.

Mathematical formula

$$ D(x,y)=G(x,y,k\sigma)-G(x,y,\sigma) $$

The response for image I is:

$$ D*I=[G_{k\sigma}-G_\sigma]*I $$

where σ is the smaller Gaussian scale and $k\sigma$ is the larger scale.

Block diagram

Input Image

↓

Gaussian 1

↓

Small σ

↓

Gaussian 2

↓

Large σ

↓

Subtract the Two Blurred Images

↓

Difference Image / Feature Response

Working

1. Take the input image.
2. Blur it using a Gaussian filter with a small standard deviation.
3. Blur the same image using a larger standard deviation.
4. Subtract one blurred image from the other.

5. Identify strong positive and negative responses or local extrema, depending on the application.
6. Use the detected structures as features or edge-related information.

Algorithm

Input: Image I, scales σ and kσ

1. A = GaussianBlur(I, σ)

2. B = GaussianBlur(I, kσ)

3. D = A - B

4. Detect significant responses in D.

5. Return the response map or detected features.

Advantages

- Computationally efficient.
- Approximates LoG without directly calculating second derivatives.
- Useful for detecting features at different scales.
- Forms the basis of the scale-space extrema detection stage in SIFT.

Disadvantages

- Choice of scales affects the result.
- Small features can be lost through excessive smoothing.
- Noise can generate unwanted responses.
- It is an approximation to LoG, not exactly the same operation.

Applications

SIFT keypoint detection, blob detection, texture analysis, image enhancement, and multi-scale image processing.

6. Comparison of Canny, LoG and DoG

Feature	**Canny**	**LoG**	**DoG**
Basic principle	Gradient and hysteresis	Gaussian + Laplacian	Difference of two Gaussians

Derivative	First derivative	Second derivative	Approximation to second-derivative response
Edge selection	Non-maximum suppression and thresholds	Zero crossings	Response changes or extrema
Noise handling	Gaussian smoothing	Gaussian smoothing	Gaussian smoothing at two scales
Edge thickness	Usually thin	Can produce paired responses	Depends on detection method
Computation	Multiple stages	Filtering and zero-crossing analysis	Two smoothing operations and subtraction
Main strength	Well-localized edges	Detects edges in all orientations	Efficient multi-scale feature detection
Main limitation	Threshold selection	Sensitive zero crossings	Scale selection

Exam point: Canny is generally preferred when clean, thin edges are required. LoG is useful when second-derivative zero crossings are needed. DoG is especially useful for scale-space feature detection and approximating LoG efficiently.

7. Line Detection — Hough Transform

Definition

The Hough Transform is a feature detection technique used to detect geometric shapes such as straight lines and circles in an image, even when the shapes are partially broken or affected by noise.

It transforms points in image space into a parameter space, where votes are accumulated to identify likely shapes.

Principle

A straight line can be represented using slope-intercept form:

$$ y=mx+c $$

However, this representation cannot describe vertical lines conveniently. The normal form is therefore preferred:

$$ \rho=x\cos\theta+y\sin\theta $$

where:

- \(\rho\) is the perpendicular distance from the origin to the line.
- \(\theta\) is the angle of the line's normal.
- \((x,y)\) is a point on the image.

Every edge point votes for possible lines in parameter space. Peaks in the accumulator indicate candidate lines.

Block diagram

Input Image

↓

Edge Detection (e.g., Canny)

↓

Initialize Hough Accumulator

↓

Each Edge Pixel Votes for Possible Lines

↓

Find Peaks in Accumulator

↓

Detected Lines on Image

Working

1. Convert the image to grayscale.
2. Detect edges, usually with Canny.
3. Create an accumulator array for possible \((\rho,\theta)\) values.
4. For every edge pixel, calculate possible line parameters.
5. Increment the corresponding accumulator cells.
6. Find peaks above a selected threshold.
7. Convert the detected parameters into line segments in the image.

Algorithm

Input: Image I

1. E = Canny(I)

2. Initialize accumulator A(ρ, θ) = 0

3. For each edge point (x, y):

    For each allowed angle θ:

        ρ = x cos θ + y sin θ

Increment A(ρ, θ)

4. Find local peaks in A.

5. Convert peaks into detected lines.

Output: Image with detected lines

Advantages

- Detects lines despite gaps in edges.
- Tolerates some noise and incomplete boundaries.
- Can detect multiple lines.
- Can be extended to circles and other parameterized shapes.

Disadvantages

- Accumulator memory increases with parameter resolution.
- Computational cost can be high for large images.
- Threshold selection affects detection.
- Basic Hough Transform detects infinite lines; additional processing is needed to obtain finite segments.

Applications

Road-lane detection, document analysis, industrial inspection, building-edge extraction, and geometric shape detection.

Circular Hough Transform

A circle is represented by:

\[(x-a)^2+(y-b)^2=r^2 \]

Here, \((a,b)\) is the centre and \(r\) is the radius. The algorithm accumulates votes for candidate centres and radii.

It is useful for detecting coins, circular components, pupils, and round industrial objects.

8. Corner Detection — Harris and Hessian-Based Methods

8.1 Harris Corner Detector

Definition: Harris corner detection identifies points where image intensity changes significantly in two different directions. Such points are useful for matching and tracking.

A corner typically has substantial gradients in both horizontal and vertical directions.

Mathematical principle

The local structure matrix is:

$$ M= \sum_{(x,y)\in W} w(x,y) \begin{bmatrix} I_x^2&I_xI_y\\ I_xI_y&I_y^2 \end{bmatrix} $$

The Harris response is:

$$ R=\det(M)-k[\operatorname{trace}(M)]^2 $$

where k is typically an empirical constant and W is a local image window.

Interpretation:

- Large positive R: likely corner.
- Large negative R: likely edge.
- Small $|R|$: likely flat region.

Block diagram

Input Image

↓

Compute Horizontal and Vertical Gradients

↓

Calculate Local Structure Matrix M

↓

Compute Harris Response R

↓

Threshold the Response

↓

Non-Maximum Suppression

↓

Corner Points

Working

1. Calculate image gradients I_x and I_y.
2. Compute the products I_x^2, I_y^2, and I_xI_y.
3. Smooth these products over a local window.
4. Calculate the Harris response R.
5. Threshold the response.
6. Apply non-maximum suppression to retain distinct corners.

Advantages

- Relatively simple and efficient.

- Works in different edge orientations.
- Useful for feature matching and tracking.
- Provides repeatable corners under moderate image changes.

Disadvantages

- Not inherently scale invariant.
- Can be affected by noise and illumination changes.
- Requires threshold and window-size selection.

Applications

Image registration, panorama stitching, optical-flow tracking, motion estimation, and object recognition.

8.2 Hessian-Based Feature Detection

The Hessian matrix uses second-order derivatives:

\[H= \begin{bmatrix} I_{xx}&I_{xy}\\ I_{xy}&I_{yy} \end{bmatrix} \]

Its determinant is:

\[\det(H)=I_{xx}I_{yy}-I_{xy}^{2} \]

Large responses can indicate blob-like structures and distinctive local features. Hessian-based methods are also used in blob detection and SURF.

Difference: Harris primarily uses products of first-order gradients, whereas Hessian-based methods use second-order derivatives to characterize local image curvature.

9. Histogram of Oriented Gradients (HOG)

Definition

Histogram of Oriented Gradients is a feature descriptor that represents the distribution of local gradient directions in an image. It captures object shape and edge structure and is widely associated with pedestrian detection.

Working principle

HOG divides an image into small cells, calculates gradient orientations, builds histograms for the cells, and normalizes groups of histograms to improve robustness to illumination and contrast.

Block diagram

Input Image

Gradient Calculation

↓

Divide Image into Cells

↓

Build Orientation Histograms

↓

Normalize Blocks

↓

Concatenate HOG Feature Vector

↓

Classifier (e.g., SVM)

Algorithm

1. Convert the image to grayscale if appropriate.
2. Calculate horizontal and vertical gradients.
3. Compute gradient magnitude and orientation.
4. Divide the image into cells.
5. Accumulate weighted orientation histograms.
6. Group neighbouring cells into blocks and normalize.
7. Concatenate the normalized histograms into a feature vector.
8. Supply the vector to a classifier.

Gradient magnitude:

$$ m=\sqrt{G_x^2+G_y^2} $$

Gradient orientation:

$$ \theta=\tan^{-1}(G_y/G_x) $$

Advantages

- Represents shape and edge patterns effectively.
- Relatively robust to small illumination and contrast changes after normalization.
- Works well with classifiers such as SVM.
- Does not require deep neural networks.

Disadvantages

- Sensitive to large rotations and scale changes unless additional techniques are used.
- Can be computationally expensive over dense sliding windows.
- Performance may degrade with complex backgrounds or occlusion.

Applications

Pedestrian detection, vehicle detection, human pose-related systems, object recognition, and visual inspection.

10. Scale-Invariant Feature Transform (SIFT)

Definition

SIFT is a local feature detection and description algorithm that identifies distinctive keypoints and generates descriptors designed to be robust to changes in image scale and rotation.

It is commonly used for matching the same object across different images.

Main stages

1. Scale-space construction: create progressively blurred and downsampled images.
2. Extrema detection: identify candidate keypoints using differences of Gaussian images.
3. Keypoint localization: refine candidate locations and reject unstable points.
4. Orientation assignment: assign a dominant orientation using local gradients.
5. Descriptor generation: create a descriptor based on local gradient orientations, commonly a 128-dimensional vector.

Block diagram

Input Image

↓

Gaussian Scale Space

↓

Difference-of-Gaussians Extrema

↓

Keypoint Localization

↓

Orientation Assignment

↓

128-Dimensional Descriptor

↓

Feature Matching

Advantages

- Strong robustness to scale and rotation changes.
- Distinctive local descriptors.
- Can match features under moderate illumination and viewpoint variation.

- Works with partially occluded objects if sufficient keypoints remain.

Disadvantages

- More computationally expensive than simpler descriptors.
- Requires storage for many keypoints and descriptors.
- Performance can decline under severe viewpoint or lighting changes.
- May produce too many features in textured scenes.

Applications

Panorama stitching, object recognition, image registration, 3D reconstruction, and visual localization.

11. SURF — Speeded-Up Robust Features

Definition

SURF is a local feature detection and description method designed to provide scale and rotation robustness while reducing computational cost compared with the original SIFT approach.

Working

1. Construct a scale-space representation.
2. Use approximations to Hessian-based responses for keypoint detection.
3. Use box filters and integral images to speed up computation.
4. Estimate dominant orientation.
5. Build a descriptor from local Haar-wavelet responses.
6. Match descriptors between images.

Block diagram

Input Image

Integral Image

Approximate Hessian Response

Keypoint Detection

Orientation Estimation

SURF Descriptor

Descriptor Matching

Advantages

- Uses integral images for efficient filtering.
- Designed to be robust to scale and rotation.
- Often faster than conventional SIFT implementations.
- Useful for matching distinctive local structures.

Disadvantages

- Can be less distinctive than SIFT in some scenes.
- Sensitive to severe viewpoint and illumination changes.
- Availability may depend on the OpenCV build and contributed modules.
- Its patented history and licensing have limited use in some environments.

Applications

Object tracking, image registration, panorama creation, and visual matching.

12. SIFT vs. SURF vs. HOG

Feature	**SIFT**	**SURF**	**HOG**
Main purpose	Local keypoint matching	Fast local keypoint matching	Shape and gradient description
Keypoints	Yes	Yes	Usually dense cells
Scale robustness	Strong	Strong	Limited unless multi-scale
Rotation robustness	Strong	Strong	Limited in standard form
Descriptor	Commonly 128 dimensions	Commonly 64 or 128 dimensions	Depends on cell/block configuration
Typical use	Matching the same object	Faster local matching	Pedestrian/object classification

13. GLOH — Gradient Location and Orientation Histogram

Definition

Gradient Location and Orientation Histogram (GLOH) is a local feature descriptor that extends the idea of SIFT-style gradient histograms by using a log-polar spatial arrangement around a keypoint.

It represents both gradient orientation and the spatial distribution of image gradients.

Working

1. Detect a local keypoint.
2. Select a neighbourhood around it.
3. Divide the neighbourhood into spatial bins using a log-polar layout.
4. Calculate gradient orientation histograms within the bins.
5. Concatenate the histograms.
6. Reduce descriptor dimensionality using PCA in the original approach.
7. Match descriptors using a distance measure.

Block diagram

Image and Keypoint Detection

Extract Local Patch

Log-Polar Spatial Binning

Gradient Orientation Histograms

Descriptor Vector

Optional PCA Reduction

Feature Matching

Advantages

- Captures spatial arrangement as well as gradient orientation.
- Can provide distinctive descriptors.
- Useful for matching under moderate image changes.

Disadvantages

- More complex than simple histogram descriptors.
- Requires keypoint detection and descriptor matching.
- Computational cost can be significant.
- Robustness still depends on image quality and viewpoint.

Applications

Image matching, object recognition, image registration, and visual search.

14. Scale-Space Analysis

Definition

Scale-space analysis studies an image at multiple levels of detail by progressively smoothing it with Gaussian filters. It helps computer vision algorithms detect features of different sizes.

Large-scale representations emphasize broad structures, while fine-scale representations preserve smaller details.

Mathematical representation

$$ L(x,y,\sigma)=G(x,y,\sigma)*I(x,y) $$

where I is the original image and σ controls the scale.

Working

1. Start with an image.
2. Apply Gaussian smoothing at several scales.
3. Compare image structures between scales.
4. Identify features that remain meaningful across scales.
5. Use these features for detection, matching, or segmentation.

Block diagram

Original Image

Gaussian Scale-Space Construction

Fine Scale

Medium Scale

Coarse Scale

Multi-Scale Feature Detection

Advantages

- Detects features of different sizes.
- Supports scale-robust feature detection.
- Separates fine details from broad structures.
- Forms the foundation of several feature detection algorithms.

Disadvantages

- Requires multiple image representations.
- Increases computation and memory requirements.
- Small details may disappear at coarse scales.

Applications

SIFT, blob detection, image pyramids, medical image analysis, and object detection.

15. Image Pyramids

Definition

An image pyramid is a sequence of representations of an image at different resolutions. It allows algorithms to analyze both coarse structures and fine details efficiently.

Types

Gaussian pyramid: Each level is produced by smoothing and downsampling the previous level.

Laplacian pyramid: Each level represents the difference between a Gaussian pyramid level and an expanded version of the next coarser level. It preserves detail information useful for reconstruction and blending.

Block diagram

Level 0 — Original Image

Level 1 — Half Resolution

Level 2 — Quarter Resolution

Coarse Levels

Gaussian pyramid algorithm

1. Set the original image as level zero.
2. Apply Gaussian smoothing.
3. Downsample the filtered image.
4. Store it as the next pyramid level.
5. Repeat until the desired number of levels is reached.

Advantages

- Reduces processing cost at coarse resolutions.

- Helps detect objects of different sizes.
- Useful for coarse-to-fine optical flow and image registration.

Disadvantages

- Downsampling loses fine details.
- Memory is needed to store multiple levels.
- Results depend on smoothing and scale choices.

Applications

Object detection, optical flow, image blending, multi-scale segmentation, and motion estimation.

16. Gaussian Derivative Filters

Definition

Gaussian derivative filters calculate image gradients by differentiating a Gaussian-smoothed image. They combine noise reduction with derivative estimation.

Mathematical formula

First-order derivatives:

$$ I_x=\frac{\partial G}{\partial x}*I,\qquad I_y=\frac{\partial G}{\partial y}*I $$

Second-order derivatives:

$$ I_{xx}=\frac{\partial^2G}{\partial x^2}*I $$

$$ I_{yy}=\frac{\partial^2G}{\partial y^2}*I $$

Working

1. Select a Gaussian scale.
2. Calculate the required derivative of the Gaussian kernel.
3. Convolve the derivative kernel with the image.
4. Use the response for edge, corner, or blob analysis.

Advantages

- Smooths noise while calculating derivatives.
- Supports first- and second-order analysis.
- Works in multiple directions.

Disadvantages

- Large smoothing scales blur small details.
- Requires choosing an appropriate scale.
- Filtering adds computation.

Applications

Canny detection, scale-space analysis, corner detection, blob detection, and image segmentation.

17. Gabor Filters

Definition

A Gabor filter is a linear filter based on a sinusoidal wave multiplied by a Gaussian envelope. It analyzes image texture and local frequency content at particular orientations and scales.

It behaves similarly to a localized frequency detector.

Mathematical representation

A common 2D Gabor form is:

$$ g(x,y)=e^{-\frac{x'^2+\gamma^2y'^2}{2\sigma^2}} \cos\left(\frac{2\pi x'}{\lambda}+\psi\right) $$

where

$$ x'=x\cos\theta+y\sin\theta $$

and θ is the orientation, λ is the wavelength, σ controls spread, γ controls aspect ratio, and ψ is phase.

Block diagram

Input Image

Choose Gabor Orientations and Scales

Convolve Image with Filter Bank

Calculate Filter Responses

Combine Responses into Texture Features

Texture Classification or Segmentation

Working

1. Choose filter scales and orientations.
2. Construct a bank of Gabor filters.

3. Convolve each filter with the image.
4. Calculate response magnitude or energy.
5. Combine responses into a feature representation.
6. Classify or segment regions based on the texture responses.

Advantages

- Effective for texture representation.
- Captures local spatial and frequency information.
- Can detect directional patterns.
- Useful for repetitive textures and surface analysis.

Disadvantages

- Many filters may be required.
- Computation can be expensive.
- Filter parameters must be selected carefully.
- Illumination changes can influence response strength.

Applications

Fingerprint recognition, face analysis, texture segmentation, fabric inspection, and medical imaging.

18. DWT — Discrete Wavelet Transform

Definition

The Discrete Wavelet Transform decomposes an image into components representing different spatial frequencies and resolutions. It separates approximation information from horizontal, vertical, and diagonal detail information.

Components in 2D DWT

- LL: Low-frequency approximation.
- LH: Detail primarily associated with one direction.
- HL: Detail primarily associated with the perpendicular direction.
- HH: Diagonal detail.

The exact directional naming depends on the filtering convention.

Block diagram

Input Image

Row and Column Low-Pass / High-Pass Filtering

LL

Approximation

LH

Detail

HL

Detail

HH

Diagonal Detail

Working

1. Apply low-pass and high-pass filters along image rows.
2. Downsample the results.
3. Apply low-pass and high-pass filters along columns.
4. Downsample again.
5. Obtain the LL, LH, HL, and HH sub-bands.
6. Optionally repeat the decomposition on the LL sub-band for multi-level analysis.

Advantages

- Provides spatial and frequency information together.
- Supports multi-resolution analysis.
- Useful for image compression and denoising.
- Separates approximation and detail coefficients.

Disadvantages

- Choice of wavelet affects results.
- Downsampling can affect shift invariance.
- Boundary handling can influence coefficients.
- Multi-level processing can increase implementation complexity.

Applications

Image compression, denoising, texture analysis, feature extraction, medical imaging, and image fusion.